2/

(a) $f(x) = \begin{cases} 1 & , -1 < x < 1 \\ 0 & , \text{otherwise} \end{cases}$

$$A(w) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(x)\cos(ws)\,ds \quad , \quad B(w) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(s)\sin(ws)\,ds$$

$$A(w) = \frac{1}{\pi}\int_{-1}^{1} \cos(ws)\,ds \quad , \quad B(w) = \frac{1}{\pi}\int_{-1}^{1} \sin(ws)\,ds$$

$$= \frac{1}{\pi}\left[\frac{1}{w}\sin(ws)\right]_{-1}^{1} \qquad = \frac{1}{\pi}\left[-\frac{1}{w}\cos(ws)\right]_{-1}^{1}$$

$$= \frac{1}{w\pi}\left(\sin(w) - \sin(-w)\right) \qquad = \frac{1}{w\pi}\left(\cos(w) - \cos(-w)\right)$$

$$= \frac{2\sin(w)}{w\pi} \qquad = 0$$

$$\boxed{f(x) = \int_0^\infty \frac{2\sin(w)}{w\pi} \cos(wx)\, dw}$$

(b)

$$f(x) = \begin{cases} \sin(x), & 0 < x < \pi \\ 0, & \text{otherwise} \end{cases}$$

$$A(w) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(x)\cos(ws)\, ds$$

$$= \frac{1}{\pi}\int_0^{\pi} \sin(s)\cos(ws)\, ds$$

$$= \frac{1}{\pi}\int_0^{\pi} \frac{1}{2}\left[\sin((1+w)s) + \sin((1-w)s)\right] ds$$

$$= \frac{1}{2\pi}\left[\frac{-\cos((1+w)s)}{1+w} + \frac{-\cos((1-w)s)}{1-w}\right]\Bigg|_0^{\pi}$$

$$= \frac{1}{2\pi}\left[\frac{-\cos((1+w)\pi)}{1+w} + \frac{1}{1+w} - \frac{\cos((1-w)\pi)}{1-w} + \frac{1}{1-w}\right]$$

$$= \frac{1}{2\pi}\left[\frac{-(1-w)\cos((1+w)\pi)}{1-w^2} + \frac{1-w}{1-w^2} - \frac{(1+w)\cos((1-w)\pi)}{1-w^2} + \frac{1+w}{1-w^2}\right]$$

$$= \frac{1}{2\pi}\left[\frac{2}{1-w^2} - \frac{\cos((1+w)\pi)}{1-w^2} + \frac{w\cos((1+w)\pi)}{1-w^2} - \frac{\cos((1-w)\pi)}{1-w^2} - \frac{w\cos((1-w)\pi)}{1-w^2}\right]$$

$$= \frac{1}{2\pi}\left[\frac{2}{1-w^2} + \frac{\cos(w\pi)}{1-w^2} - \cancel{\frac{w\cos(w\pi)}{1-w^2}} + \frac{\cos(w\pi)}{1-w^2} + \cancel{\frac{w\cos(w\pi)}{1-w^2}}\right]$$

$$= \frac{1}{2\pi}\left[\frac{2}{1-w^2} + \frac{2\cos(w\pi)}{1-w^2}\right] = \frac{1+\cos(w\pi)}{\pi(1-w^2)}$$

$$B(w) = \frac{1}{\pi}\int_{-\infty}^{\infty} f(x)\sin(ws)\, ds$$

$$= \frac{1}{\pi}\int_0^{\pi} \sin(s)\sin(ws)\, ds$$

$$= 0 \text{ for } w \neq 1, \quad = \frac{\pi}{2}\cdot\frac{1}{\pi} \text{ for } w = 1$$

$$= \frac{1}{2} \text{ for } w = 1$$

$$= \frac{1}{2}\delta(w-1)$$

$$\boxed{f(x) = \int_0^\infty \left[\left(\frac{1+\cos(w\pi)}{\pi(1-w^2)} \right) \cos(wx) + \frac{1}{2}\,\delta(w-1)\,\sin(wx) \right] dw}$$